

PAPER_173: Modular Compressed MUGE — 9-Term Mathematical Decomposition

Abstract

This paper presents a UQFF analysis of astrophysical observables, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

Whitepaper §2.4-E | Thread 381a8fe7 | Session 48

Abstract

The Modular Unified Gravity Equation (MUGE) in compressed form decomposes gravitational dynamics into 9 independent sub-terms. Each term captures a distinct physical contribution: Newtonian base, cosmological expansion, magnetic suppression, environmental context, Ug contributions, cosmological constant, quantum corrections, fluid dynamics, and density perturbations.

UQFF Discovery: Novel application of UQFF calibration constants ($\kappa = 5.0 \times 10^{-4} \text{ day}^{-1}$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

$$F_U(r, t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{b_i}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$g_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \left(1 - [SSq] \cdot U_{b_i} / F_U(r, t)\right), \quad [SSq] = 0.57$$

1. MUGESystem Struct

```
struct MUGESystem {
    std::string name;
    double I;           // Moment of inertia [kg·m²]
    double A;           // Area / cross-section [m²]
    double omega1, omega2; // Angular frequencies [rad/s]
    double Vsys;        // System volume [m³]
    double vexp;        // Expansion velocity [m/s]
    double t;           // System age [s]
    double z;           // Redshift
    double ffluid;      // Fluid frequency [Hz]
    double M;           // System mass [kg]
    double r;           // Characteristic radius [m]
    double B, Bcrit;    // Magnetic field and critical field [T]
    double rho_fluid;   // Fluid density [kg/m³]
    double g_local;     // Local gravitational acceleration [m/s²]
```

```

double M_DM;           // Dark matter mass [kg]
double delta_rho_rho; // Density perturbation ratio
};

```

2. Nine Sub-Terms

Term 1 — Newtonian Gravitational Baseline

$\text{compressed_base} = G \times M / r^2$
 Constants: $G = 6.67430 \times 10^{-11}$
 Test: $M = 1.989 \times 10^{30} \text{ kg}$, $r = 1.496 \times 10^{11} \text{ m}$? $\sim 0.0059 \text{ m/s}^2$

Term 2 — Hubble Expansion

$\text{compressed_expansion} = 1 + H_0 \times v_{\text{exp}}$
 $H_0 = 2.269 \times 10^{-18} \text{ s}^{-1}$ ($\sim 70.1 \text{ km/s/Mpc}$)
 At $t=0$: $\text{expansion} = 1.0$

Term 3 — Magnetic Suppression (Super Adjustment)

$\text{compressed_super_adj} = 1 - B/B_{\text{crit}}$
 Test: $B = 1 \times 10 \text{ T}$, $B_{\text{crit}} = 1 \times 10^{11} \text{ T}$? 0.9
 Above B_{crit} : approaches 0 (magnetic quench)

Term 4 — Environmental Factor

$\text{compressed_env} = 1.0$ [placeholder for ISM/nebular environment]

Term 5 — Ug Contributions Sum

$\text{compressed_Ug_sum} = 0.0$ [Ug interface placeholder for future coupling]

Term 6 — Cosmological Constant Term

$\text{compressed_cosm} = \rho \times c^2 / 3$
 $\rho = 1.1 \times 10^{-52} \text{ m}^{-2}$ (dark energy)
 $c = 3 \times 10^8 \text{ m/s}$
 ? $\text{compressed_cosm} = 1.1 \times 10^{-52} \times 9 \times 10^{16} / 3 \sim 3.3 \times 10^{-37} \text{ m/s}^2$

Term 7 — Quantum Correction

$\text{compressed_quantum} = (\hbar / \Delta x_p) \times \Delta E \times (2p / t_{\text{Hubble}})$

Parameters:

$\hbar$	$= 1.0546 \times 10^{-34} \text{ J}\cdot\text{s}$
Δx_p	$= \Delta x \times \Delta p = 1 \times 10^{-68} \text{ J}\cdot\text{m}$ (minimal uncertainty product)
ΔE	$= \text{integral_psi} = 2.176 \times 10^{-18}$ (ground state energy proxy)
t_{Hubble}	$= 4.35 \times 10^{17} \text{ s}$

? quantum = (1.0546e-34 / 1e-68) × 2.176e-18 × (2p / 4.35e17)
 = 1.0546e34 × 2.176e-18 × 1.443e-17
 ~ 3.312e-1

Term 8 — Fluid Dynamics

compressed_fluid = rho_fluid × Vsys × g_local
 Test (SGR1745): rho_fluid=1e-15, Vsys=4.189e12, g_local=10.0
 ? compressed_fluid = 1e-15 × 4.189e12 × 10 = 4.189e-2

Term 9 — Density Perturbation

compressed_perturbation = M × (delta_rho_rho + 3 × G × M / r³)
 Captures dark matter and baryonic density contrast effects.

3. Full Compressed MUGE

compressed_MUGE = base × expansion × super_adj × env × (1 + Ug_sum)
 + cosm + quantum + fluid + perturbation

Expected (SGR1745):
 base ~ G×2.984e30/1e8 ~ 1.99e11
 expansion ~ 1 + 2.269e-18×1e3 ~ 1.0
 fluid = 4.189e-2
 perturbation ~ M×d? terms
 Total ? ~ 1.782e39 (from unit test)

4. Canonical System Test Values

System	compressed_MUGE [m/s ²]
SGR 1745-2900	~ 1.782e39
Sagittarius A*	~ 1.782e39×(M_SgrA/M_SGR)
Student Guide	cosmological scale

5. Relationship to SOURCE4

The 9-term compressed MUGE directly corresponds to the compute_compressed_MUGE_SOURCE4() function in MAIN_1_CoAnQi.cpp (lines 25623–26026, namespace SOURCE4). The thread 381a8fe7 version provides the modular sub-term decomposition enabling independent validation.

6. References

- MUGE.h/cpp (thread 381a8fe7)
- UnitTests.cpp lines 1-200 (test_compute_compressed_MUGE expected=1.782e39)
- PAPER_174 (resonance MUGE, same MUGESystem struct)
- SOURCE4 integration commit 3e66d94